COMPUTER NETWORKS
BTEUP — 3rd Semester CS
Exam-Ready Notes (60 Marks Pattern)
Developer:- Vipin Gangwar Owner
For More Join :- Telegram

PAPER PATTERN:
• Q1 to Q5: "Attempt any two" → 2 × 5 = 10 marks each
• Q6: Short notes on any four → 4 × 2.5 = 10 marks
• Total: 60 Marks

Q1. ATTEMPT ANY TWO:

(a) What is a Computer Network? Explain its types (LAN, MAN, WAN). [5 Marks]

Answer:
Computer Network ek aise computers ka group hai jo aapas mein connected hote hain aur data/resource share karte hain. Network ka main goal hai — communication, resource sharing, aur data exchange.

Types of Computer Networks:

1. LAN (Local Area Network):

- Chhote area mein — ek building, office, school.
- Speed bahut fast (100 Mbps – 10 Gbps).
- Cost kam hota hai.
- Example: College computer lab.

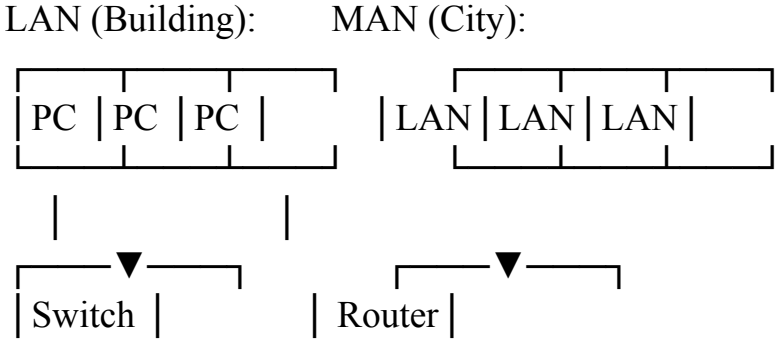
2. MAN (Metropolitan Area Network):
- Ek city ya bade campus mein faila hota hai.
- LAN se bada, WAN se chhota.
- Speed medium hoti hai.
- Example: Cable TV network, city-wide ISP.

3. WAN (Wide Area Network):
- Poore country ya world mein faila hota hai.
- Speed slow ho sakti hai.
- Cost zyada hota hai.
- Example: Internet, ATM network.

Comparison:

Feature	LAN	MAN	WAN
Area	Building	City	Country/World
Speed	Very High	Medium	Low
Cost	Low	Medium	High
Example	Office	Cable TV	Internet

[DIAGRAM: LAN, MAN, WAN]





WAN (World):



(b) Explain OSI Reference Model with diagram. [5 Marks]

Answer:
OSI (Open Systems Interconnection) Model ek 7-layer architecture hai jo network communication ko define karta hai. Har layer ka apna specific kaam hota hai.

7 Layers of OSI Model:

1. Physical Layer (Layer 1):
   - Actual hardware transmission.
   - Bits (0 aur 1) transfer karta hai.
   - Devices: Hub, Repeater, Cables.

2. Data Link Layer (Layer 2):
   - Error-free data transfer between nodes.
   - Framing, Error Detection, Flow Control.
   - Devices: Switch, Bridge.

3. Network Layer (Layer 3):
   - Routing aur Logical Addressing (IP).
   - Path selection karta hai.
   - Devices: Router.

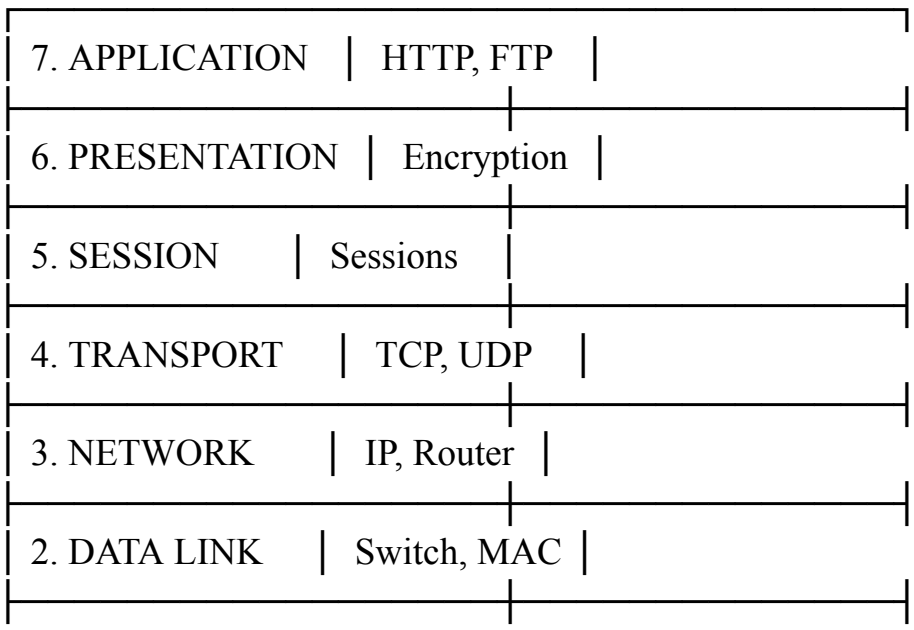
4. Transport Layer (Layer 4):
   - End-to-end communication.
   - Segmentation, Flow Control, Error Control.
   - Protocols: TCP, UDP.

5. Session Layer (Layer 5):
   - Session establish, manage, aur terminate karta hai.
   - Synchronization aur Dialog Control.

6. Presentation Layer (Layer 6):
   - Data format conversion.
   - Encryption/Decryption, Compression.
   - Example: JPEG, MPEG, ASCII.

7. Application Layer (Layer 7):
   - User interface.
   - Protocols: HTTP, FTP, SMTP, DNS.

[DIAGRAM: OSI MODEL]



1. PHYSICAL	Cables, Hub

(c) Explain TCP/IP Model with diagram. [5 Marks]

Answer:
TCP/IP Model ek practical network model hai jo Internet mein use hota hai. Ye 4 layers ka hota hai.

4 Layers of TCP/IP Model:

1. Network Access Layer (Link Layer):
   - Physical transmission.
   - OSI ke Physical + Data Link se match karta hai.
   - Protocols: Ethernet, WiFi.

2. Internet Layer:
   - Logical addressing aur routing.
   - OSI ke Network Layer se match karta hai.
   - Protocols: IP, ICMP, ARP.

3. Transport Layer:
   - End-to-end communication.
   - OSI ke Transport Layer se match karta hai.
   - Protocols: TCP, UDP.

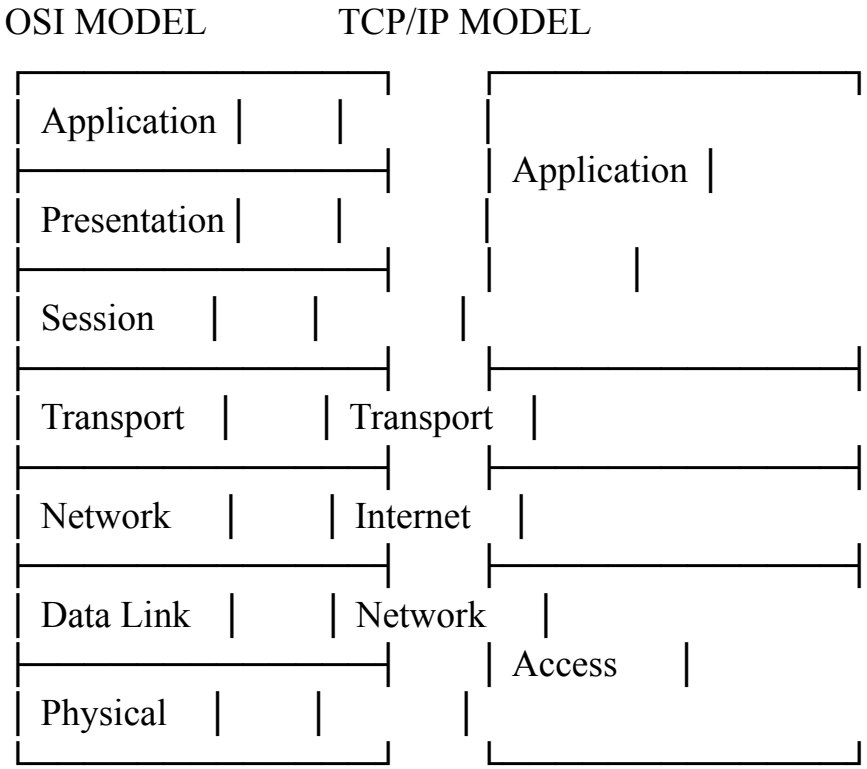
4. Application Layer:
   - User services.
   - OSI ke Session + Presentation + Application se match karta hai.

- Protocols: HTTP, FTP, SMTP, DNS.

Comparison with OSI:

OSI	TCP/IP
7 Layers	4 Layers
Theoretical	Practical
Developed first	Implemented first

[DIAGRAM: TCP/IP vs OSI]



(d) Explain Guided and Unguided Transmission Media.

[5 Marks]

Answer:
Transmission Media wo physical path hai jiske through data ek device se dusre device tak jata hai.

A. Guided Media (Wired):
Ismein data ek physical cable ke through travel karta hai.

1. Twisted Pair Cable:
- Do insulated copper wires twisted hote hain.
- Types: UTP (Unshielded), STP (Shielded)
- Speed: 100 Mbps – 1 Gbps
- Cost: Sabse sasta
- Use: Telephone, LAN

2. Coaxial Cable:
- Ek central copper conductor jiske around insulation aur shielding hoti hai.
- Speed: 10 Mbps – 100 Mbps
- Use: Cable TV, Old LANs

3. Fiber Optic Cable:
- Data light signals ke through travel karta hai.
- Speed: 1 Gbps – 100 Gbps
- Cost: Mehnga
- Use: Backbone, Internet
- Advantage: Fast, secure, long distance

B. Unguided Media (Wireless):
Ismein data air/space ke through travel karta hai.

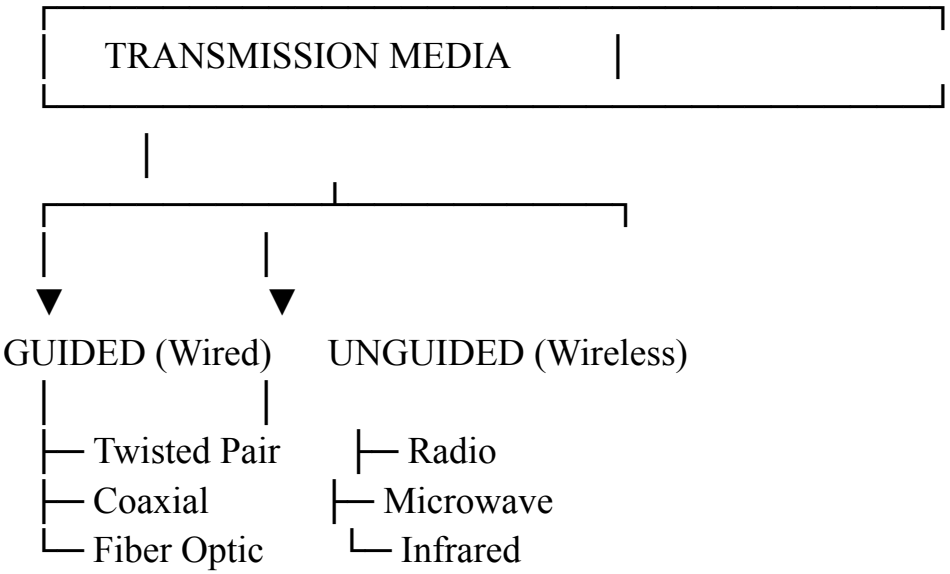
1. Radio Waves:
- Frequency: 3 KHz – 1 GHz

- Omni-directional
- Use: AM/FM Radio, TV

2. Microwaves:
- Frequency: 1 GHz – 300 GHz
- Line-of-sight communication
- Use: Satellite, Mobile

3. Infrared:
- Short distance communication
- Use: TV Remote, AC Remote

[DIAGRAM: TRANSMISSION MEDIA]



Q2. ATTEMPT ANY TWO:

(a) Explain Network Topologies with diagrams. [5 Marks]

Answer:
Network Topology wo arrangement hai jismein computers aur devices ek network mein connected hote hain.

Types of Topologies:

1. Bus Topology:
   - Sab devices ek single backbone cable se connected.
   - Simple, sasta.
   - Backbone fail → poora network down.
   - Example: Old Ethernet.

2. Star Topology:
   - Sab devices ek central hub/switch se connected.
   - Easy to manage, ek cable fail se dusre par effect nahi.
   - Central device fail → poora network down.
   - Example: Modern LAN.

3. Ring Topology:
   - Devices ek circular ring mein connected.
   - Data ek direction mein travel karta hai.
   - Token passing use hota hai.
   - Ek node fail → poora ring down.

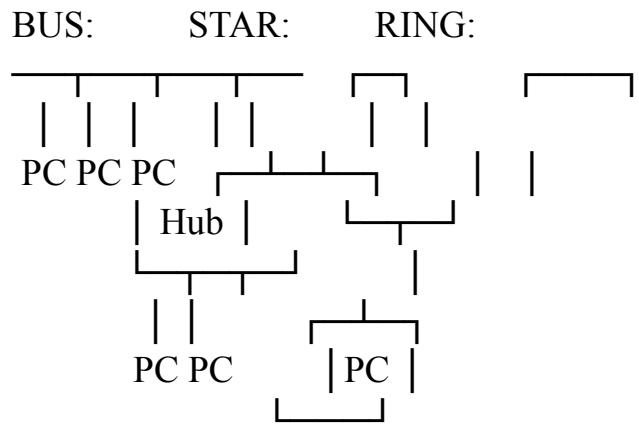
4. Mesh Topology:
   - Har device har dusre device se connected.
   - Highly reliable, secure.
   - Bahut costly, wiring complex.
   - Example: Military networks.

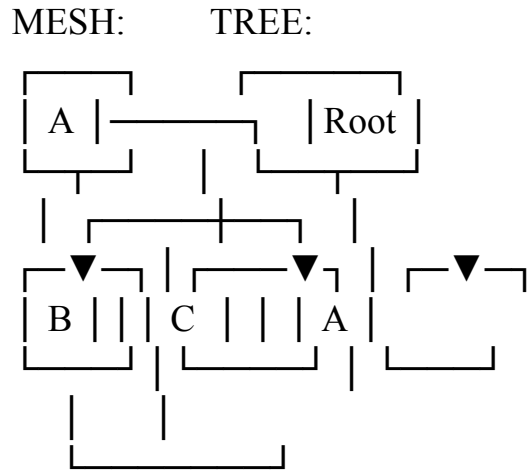
5. Tree Topology:
   - Star + Bus ka combination.

- Hierarchical structure.
- Scalable.

6. Hybrid Topology:
   - Do ya zyada topologies ka combination.

[DIAGRAM: TOPOLOGIES]



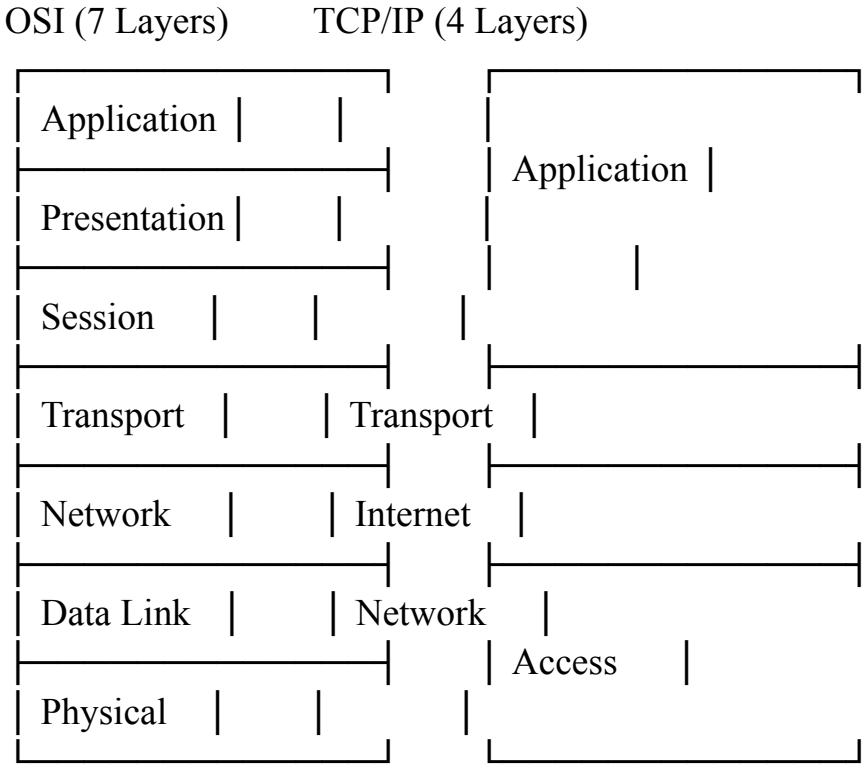


(b) Explain OSI vs TCP/IP Model difference. [5 Marks]

Answer:
OSI aur TCP/IP dono network models hain lekin dono mein kuch differences hain.

Feature	OSI Model	TCP/IP Model
Layers	7	4
Developed By	ISO	DoD (ARPA)
Type	Theoretical	Practical
Usage	Reference model	Internet
Session Layer	Separate	Application mein
Presentation	Separate	Application mein
Reliability	Both	Mostly TCP
Complexity	Complex	Simple

[DIAGRAM: COMPARISON]



(c) Explain Switching Techniques. [5 Marks]

Answer:
Switching Techniques wo methods hain jinse data ek device se dusre device tak pahunchaya jata hai.

Types of Switching:

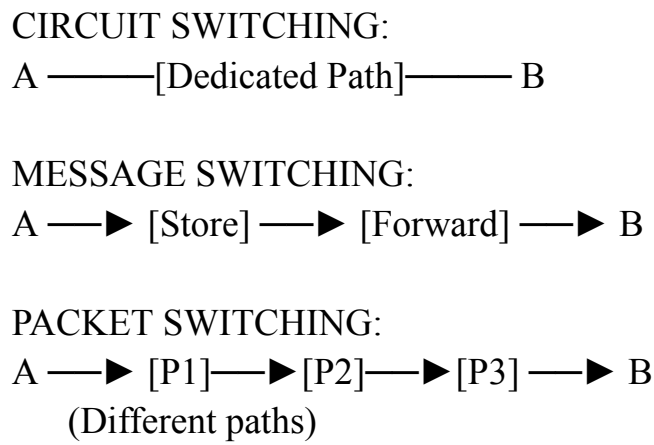
1. Circuit Switching:
   - Communication se pehle ek dedicated path establish hota hai.
   - Poora bandwidth us connection ke liye reserve hota hai.
   - Example: Traditional Telephone Network.
   - Advantage: Guaranteed bandwidth, no delay.
   - Disadvantage: Wasteful, costly.

2. Message Switching:
   - Poora message store karke aage bheja jata hai.
   - Store-and-forward technique.
   - Example: Telegraph.
   - Advantage: Efficient bandwidth use.
   - Disadvantage: Delay hota hai.

3. Packet Switching:
   - Message ko chhote packets mein divide kiya jata hai.
   - Har packet alag-alag path se ja sakta hai.
   - Example: Internet.
   - Advantage: Efficient, reliable.

- Disadvantage: Out-of-order packets possible.

[DIAGRAM: SWITCHING TECHNIQUES]



(d) Explain Network Topologies (Star, Ring, Bus).
[5 Marks]

Answer:

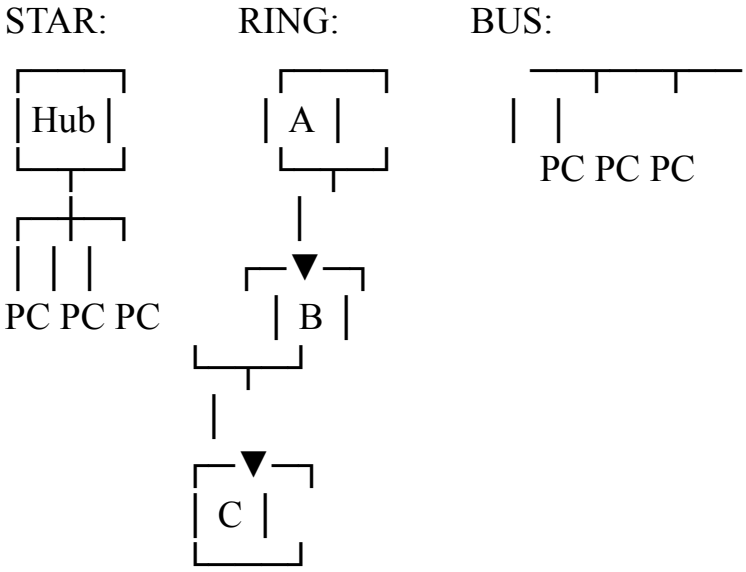
1. Star Topology:
   - Central hub/switch se sab devices connected.
   - Easy to install aur manage.
   - Ek cable fail → sirf us device par effect.
   - Central device fail → poora network down.
   - Cost: Medium.

2. Ring Topology:
   - Devices circular ring mein connected.
   - Data ek direction mein move karta hai.
   - Token passing technique.

- Ek node fail → poora network down.
- Cost: Medium.

3. Bus Topology:
- Sab devices ek backbone cable se connected.
- Simple aur sasta.
- Backbone fail → poora network down.
- Cost: Low.

[DIAGRAM: TOPOLOGY COMPARISON]



Q3. ATTEMPT ANY TWO:

(a) Explain Network Layer and IPv4 Protocol. [5 Marks]

Answer:

Network Layer OSI model ki Layer 3 hai. Iska main kaam hai — logical addressing aur routing.

Functions of Network Layer:
1. Logical Addressing (IP Address)
2. Routing (Path Selection)
3. Fragmentation aur Reassembly
4. Error Reporting (ICMP)
5. Congestion Control

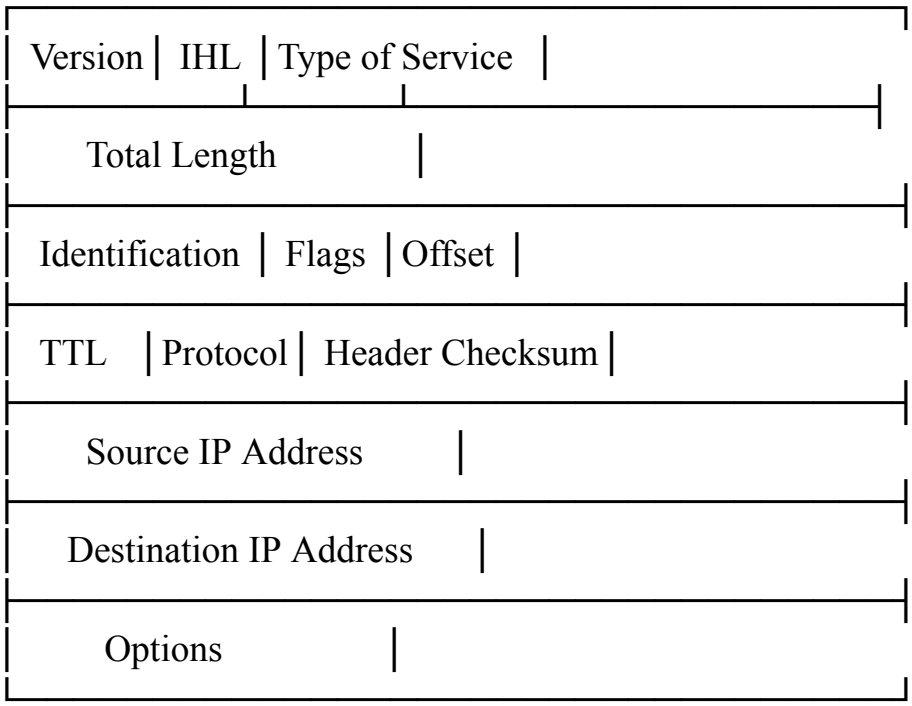
IPv4 Protocol:
- IPv4 = Internet Protocol version 4
- 32-bit address (4 bytes)
- Total 4.3 billion unique addresses
- Format: Dotted Decimal (e.g., 192.168.1.1)

IPv4 Header Fields:
1. Version (4 bits)
2. Header Length (4 bits)
3. Type of Service (8 bits)
4. Total Length (16 bits)
5. Identification (16 bits)
6. Flags (3 bits)
7. Fragment Offset (13 bits)
8. TTL (Time To Live)
9. Protocol
10. Header Checksum
11. Source Address (32 bits)
12. Destination Address (32 bits)

IPv4 Address Classes:
- Class A: 1.0.0.0 – 126.255.255.255
- Class B: 128.0.0.0 – 191.255.255.255
- Class C: 192.0.0.0 – 223.255.255.255
- Class D: Multicast

- Class E: Reserved

[DIAGRAM: IPv4 HEADER]



(b) Explain IPv6 and its advantages over IPv4.
[5 Marks]

Answer:
IPv6 (Internet Protocol version 6) IPv4 ka successor hai. Ye 128-bit address use karta hai.

IPv6 Features:
1. 128-bit address (IPv4 mein 32-bit)
2. 340 undecillion unique addresses

3. Hexa-decimal format
4. Format: 2001:0db8:85a3::8a2e:0370:7334

Advantages of IPv6 over IPv4:
1. More Addresses: IPv4 ke 4.3 billion vs IPv6 ke almost unlimited.
2. Simplified Header: Fixed header size, faster processing.
3. Built-in Security: IPSec mandatory hai.
4. No NAT Required: Har device ka unique address.
5. Better QoS: Flow labeling support.
6. Auto-Configuration: Stateless address auto-configuration.
7. No Broadcast: Multicast aur Anycast use hota hai.
8. Efficient Routing: Hierarchical addressing.

[DIAGRAM: IPv4 vs IPv6]

IPv4:	IPv6:
32-bit	128-bit
192.168.1.1	2001:0db8:85a3:: 8a2e:0370:7334
Dotted Decimal	Hexa-decimal
4.3 Billion Addresses	340 Undecillion Addresses

(c) Explain Routing Algorithms (Distance Vector, Link State). [5 Marks]

Answer:
Routing Algorithm wo method hai jisse router decide karta hai ki packet ko kaunse path se bhejna hai.

1. Distance Vector Routing:
   - Har router apne neighbors ko apna distance vector bhejta hai.
   - Distance = Hops count.
   - Bellman-Ford algorithm use hota hai.
   - Protocol: RIP
   - Advantage: Simple.
   - Disadvantage: Count-to-Infinity problem.

2. Link State Routing:
   - Har router poore network ka map banata hai.
   - Dijkstra's algorithm use hota hai.
   - Har router apne neighbors ka state janta hai.
   - Protocol: OSPF
   - Advantage: Fast convergence, no count-to-infinity.
   - Disadvantage: Complex, memory zyada.

Comparison:

Feature	Distance Vector	Link State
Algorithm	Bellman-Ford	Dijkstra
Info	Neighbors only	Full topology
Protocol	RIP	OSPF
Convergence	Slow	Fast
Memory	Low	High

[DIAGRAM: ROUTING]

DISTANCE VECTOR:

A ──1── B ──2── C
A ka distance to C = 1+2 = 3

LINK STATE:
A ──1── B ──2── C
──────────4──────────
A ka shortest path = A→B→C = 3

(d) Explain RIP and OSPF. [5 Marks]

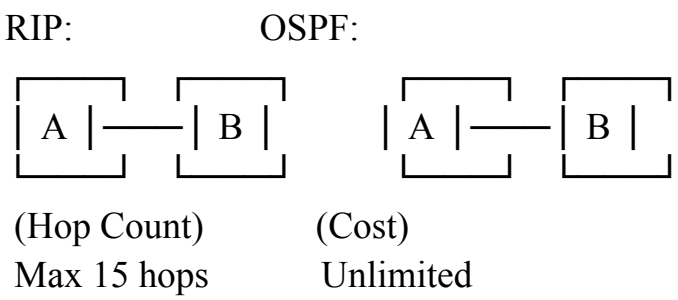
Answer:

RIP (Routing Information Protocol):
- Distance Vector routing protocol.
- Hop count metric use karta hai.
- Maximum 15 hops allowed (16 = infinity).
- Har 30 seconds mein update bhejta hai.
- Small networks ke liye suitable.
- Version: RIPv1, RIPv2
- Advantage: Simple, easy to configure.
- Disadvantage: Slow convergence, limited size.

OSPF (Open Shortest Path First):
- Link State routing protocol.
- Dijkstra's algorithm use karta hai.
- Cost metric use karta hai (bandwidth based).
- Large networks ke liye suitable.
- Areas mein divide karta hai.
- Fast convergence.
- Advantage: Scalable, fast, efficient.

- Disadvantage: Complex configuration.

[DIAGRAM: RIP vs OSPF]



Q4. ATTEMPT ANY TWO:

(a) Explain Transport Layer and its Protocols.
[5 Marks]

Answer:
Transport Layer OSI model ki Layer 4 hai. Iska main kaam hai — end-to-end communication provide karna.

Functions of Transport Layer:
1. Segmentation aur Reassembly
2. Port Addressing
3. Connection Control
4. Flow Control
5. Error Control
6. Multiplexing/Demultiplexing

Transport Layer Protocols:

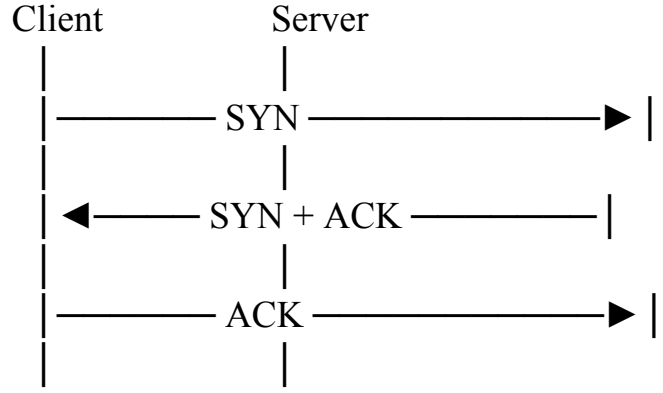
1. TCP (Transmission Control Protocol):
   - Connection-oriented protocol.
   - Reliable data transfer.
   - 3-way handshake (SYN, SYN-ACK, ACK).
   - Error checking aur retransmission.
   - Flow control (Sliding Window).
   - Use: HTTP, FTP, Email.

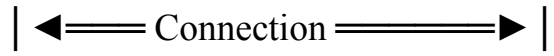
2. UDP (User Datagram Protocol):
   - Connectionless protocol.
   - Unreliable but fast.
   - No handshake, no error checking.
   - Use: Video streaming, Gaming, DNS.

Comparison:

Feature	TCP	UDP
Connection	Yes	No
Reliability	High	Low
Speed	Slow	Fast
Header	20 bytes	8 bytes
Use	HTTP, FTP	DNS, Video

[DIAGRAM: TCP 3-WAY HANDSHAKE]





(b) Explain Application Layer Protocols (HTTP, FTP, SMTP, DNS). [5 Marks]

Answer:
Application Layer OSI model ki Layer 7 hai. Ye user ko network services provide karti hai.

1. HTTP (HyperText Transfer Protocol):
   - Web pages transfer karne ke liye.
   - Port: 80 (HTTP), 443 (HTTPS)
   - Stateless protocol.
   - Methods: GET, POST, PUT, DELETE.
   - HTTPS secure version hai (SSL/TLS).

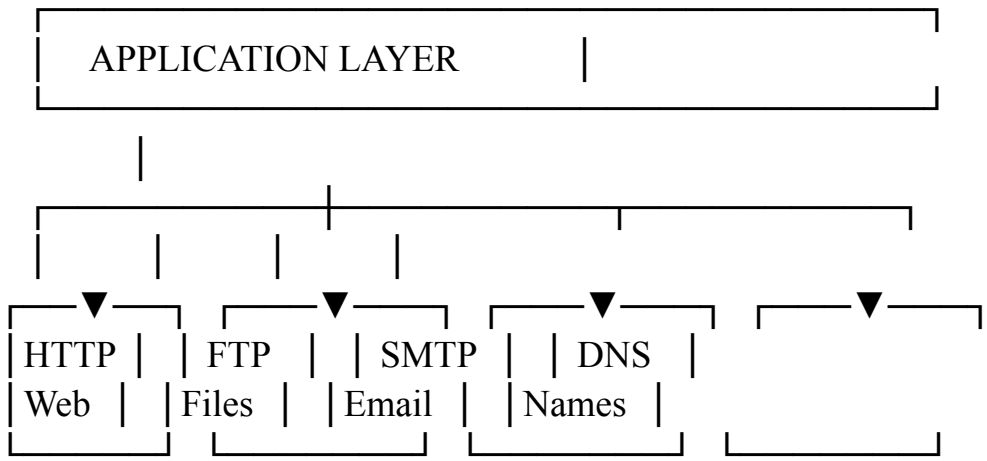
2. FTP (File Transfer Protocol):
   - Files transfer karne ke liye.
   - Port: 20 (Data), 21 (Control)
   - Two connections: Control + Data.
   - Types: Anonymous, Authenticated.

3. SMTP (Simple Mail Transfer Protocol):
   - Email bhejne ke liye.
   - Port: 25
   - Push protocol.
   - Receiving ke liye POP3/IMAP use hota hai.

4. DNS (Domain Name System):
   - Domain name ko IP address mein convert karta

hai.
- Port: 53
- Hierarchical system.
- Example: google.com → 142.250.190.46

[DIAGRAM: APPLICATION PROTOCOLS]



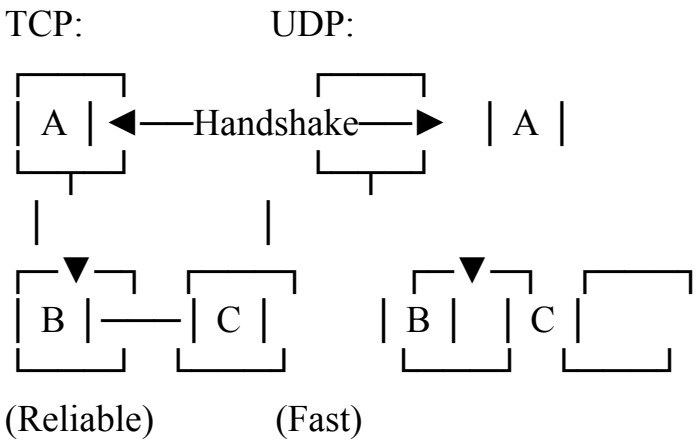
(c) Explain TCP vs UDP difference. [5 Marks]

Answer:
TCP aur UDP dono Transport Layer protocols hain
lekin dono mein bahut differences hain.

Feature	TCP	UDP
Full Form	Transmission Control Protocol	User Datagram Protocol
Connection	Connection-oriented	Connectionless
Reliability	Reliable	Unreliable
Handshake	3-way	No

Error Checking	Yes	Minimal
Flow Control	Yes	No
Speed	Slow	Fast
Header Size	20 bytes	8 bytes
Ordering	In-order	No guarantee
Use	HTTP, FTP, Email	DNS, Video, Gaming

[DIAGRAM: TCP vs UDP]



(d) Explain Ports and Sockets. [5 Marks]

Answer:

Port:
Port ek 16-bit number hai jo application layer process ko identify karta hai. Ye transport layer par use hota hai.

Port Ranges:
1. Well-Known Ports: 0 – 1023

- HTTP: 80
- FTP: 21
- SMTP: 25
- DNS: 53
- HTTPS: 443

2. Registered Ports: 1024 – 49151
   - Used by user applications.

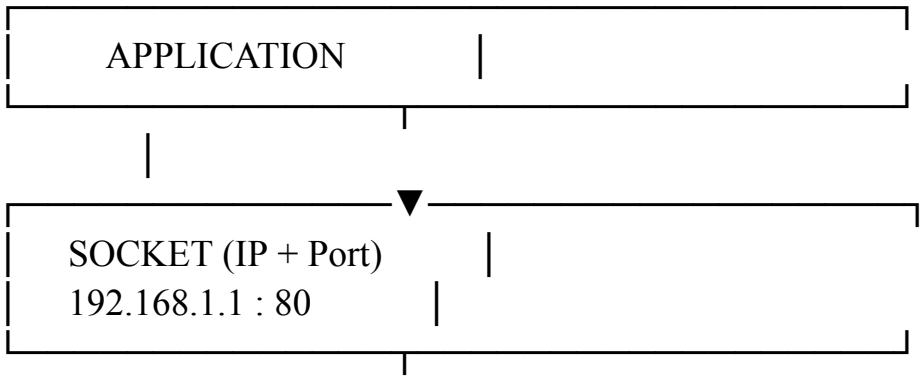
3. Dynamic Ports: 49152 – 65535
   - Temporary ports.

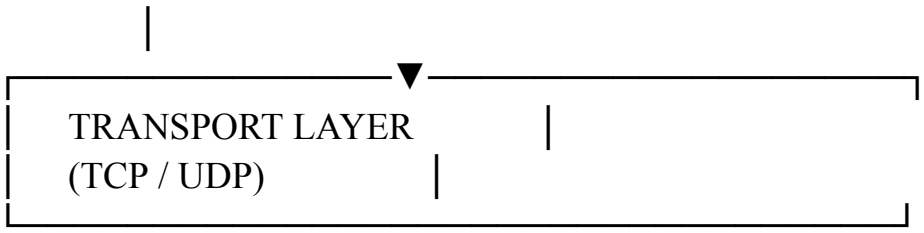
Socket:
Socket = IP Address + Port Number
Ye ek endpoint hai communication ka.

Example:
192.168.1.1:80
(IP Address : Port)

Socket Types:
1. Stream Socket (TCP)
2. Datagram Socket (UDP)
3. Raw Socket

[DIAGRAM: SOCKET]





Q5. ATTEMPT ANY TWO:

(a) Explain Network Devices (Hub, Switch, Router, Bridge, Gateway). [5 Marks]

Answer:

1. Hub:
   - Layer 1 (Physical) device.
   - Data sabhi ports par broadcast karta hai.
   - Half-duplex.
   - Collision zyada hota hai.
   - Cost: Low.

2. Switch:
   - Layer 2 (Data Link) device.
   - MAC address ke through data bhejta hai.
   - Full-duplex.
   - Collision kam.
   - Cost: Medium.

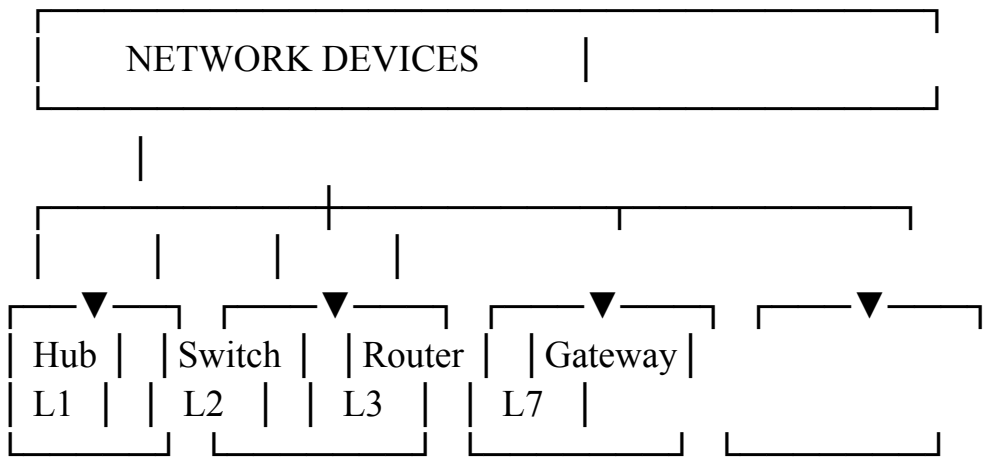
3. Router:
   - Layer 3 (Network) device.

- IP address ke through data bhejta hai.
- Different networks ko connect karta hai.
- Best path select karta hai.
- Cost: High.

4. Bridge:
- Layer 2 device.
- Do LAN segments ko connect karta hai.
- MAC address filter karta hai.
- Switch ka older version.

5. Gateway:
- Layer 7 (Application) device.
- Different protocols ko connect karta hai.
- Protocol conversion karta hai.

[DIAGRAM: NETWORK DEVICES]



(b) Explain NIC and its functions. [5 Marks]

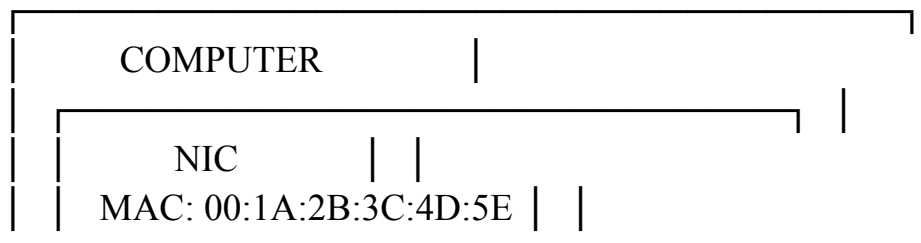
Answer:
NIC (Network Interface Card) ek hardware component hai jo computer ko network se connect karta hai.

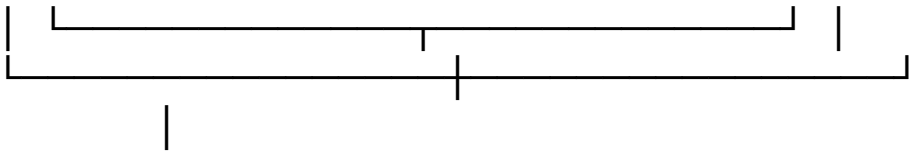
Functions of NIC:
1. Data Transmission: Data ko physical medium par bhejta hai.
2. Data Reception: Network se data receive karta hai.
3. MAC Address: Har NIC ka unique MAC address hota hai.
4. Framing: Data ko frames mein convert karta hai.
5. Error Detection: Transmission errors check karta hai.
6. Buffering: Data temporarily store karta hai.

Types of NIC:
1. Wired NIC (Ethernet)
2. Wireless NIC (WiFi)
3. Fiber NIC

MAC Address:
- 48-bit address
- Format: 00:1A:2B:3C:4D:5E
- First 24 bits: Manufacturer ID
- Last 24 bits: Device ID

[DIAGRAM: NIC]





Network Cable

(c) Explain Network Management System (NMS) and SNMP.
[5 Marks]

Answer:

NMS (Network Management System):
NMS ek software/hardware system hai jo network ko monitor, manage, aur control karta hai.

Functions of NMS:
1. Fault Management: Errors detect aur fix karna.
2. Configuration Management: Devices configure karna.
3. Performance Management: Network performance monitor karna.
4. Security Management: Security policies implement karna.
5. Accounting Management: Resource usage track karna.

SNMP (Simple Network Management Protocol):
SNMP ek application layer protocol hai jo network devices ko manage karne ke liye use hota hai.

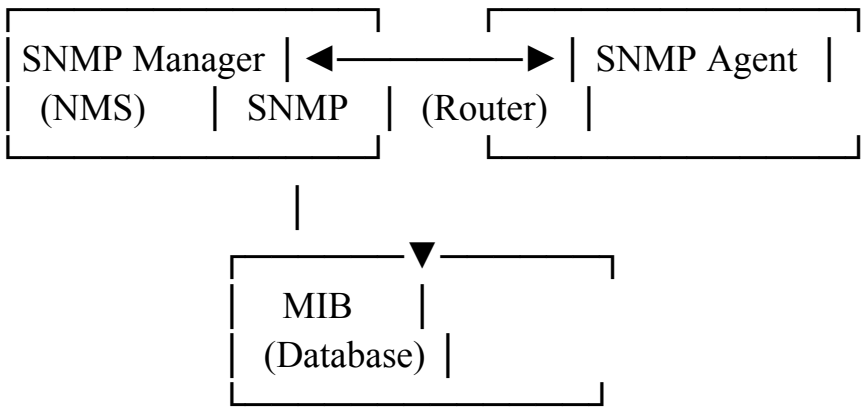
Components of SNMP:
1. SNMP Manager: Central management system.
2. SNMP Agent: Managed device par run karta hai.

3. MIB (Management Information Base): Database of managed objects.

SNMP Operations:
- GET: Value retrieve karna.
- SET: Value set karna.
- TRAP: Alert bhejna.

Port: 161 (Agent), 162 (Manager)

[DIAGRAM: SNMP]



(d) Explain WiFi Technology. [5 Marks]

Answer:
WiFi (Wireless Fidelity) ek wireless networking technology hai jo devices ko bina cable ke network se connect karti hai.

WiFi Standards:

1. 802.11a: 5 GHz, 54 Mbps
2. 802.11b: 2.4 GHz, 11 Mbps
3. 802.11g: 2.4 GHz, 54 Mbps
4. 802.11n: 2.4/5 GHz, 600 Mbps
5. 802.11ac: 5 GHz, 1 Gbps+
6. 802.11ax (WiFi 6): 10 Gbps

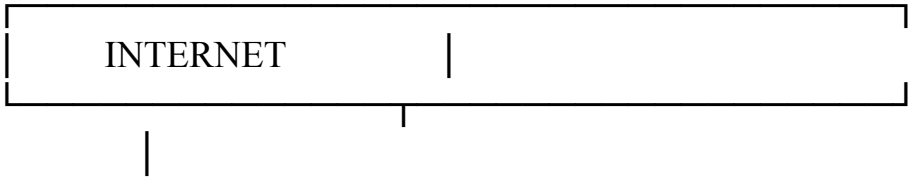
WiFi Components:
1. Access Point (AP): Central device.
2. Station: Client devices.
3. Basic Service Set (BSS): AP + Stations.
4. Extended Service Set (ESS): Multiple BSS.

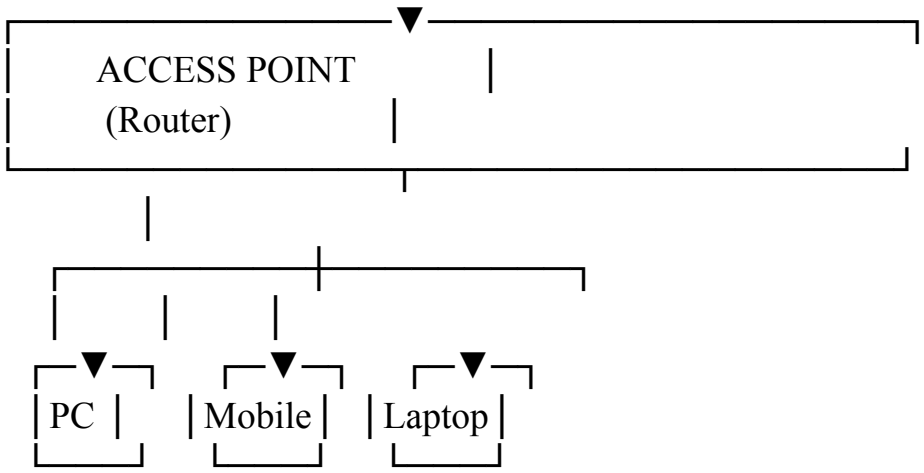
WiFi Security:
1. WEP: Weak (cracked)
2. WPA: Better
3. WPA2: Strong (AES)
4. WPA3: Strongest

Advantages:
- No cables needed
- Mobility
- Easy installation

Disadvantages:
- Limited range
- Security risks
- Interference

[DIAGRAM: WiFi NETWORK]





Q6. WRITE SHORT NOTES ON ANY FOUR:

(a) MAC Address [2.5 Marks]

Answer:
MAC (Media Access Control) address ek unique 48-bit hardware address hai jo har NIC ko assign hota hai.

Format: 00:1A:2B:3C:4D:5E (6 bytes)

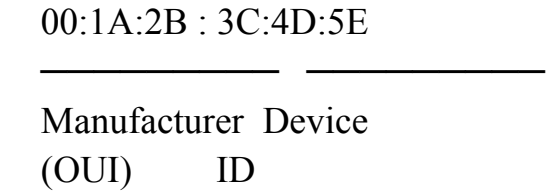
Parts:
- First 24 bits: OUI (Organizationally Unique Identifier) — Manufacturer
- Last 24 bits: Device specific

Types:
- Unicast: Single device
- Multicast: Group

- Broadcast: All devices (FF:FF:FF:FF:FF:FF)

Use: Data Link Layer par devices identify karna.

[DIAGRAM: MAC ADDRESS]



(b) IP Address Classes [2.5 Marks]

Answer:
IP Address ek 32-bit logical address hai jo network par device ko identify karta hai.

Classes:
1. Class A: 1 – 126 (Large networks)
   - Default Mask: 255.0.0.0
2. Class B: 128 – 191 (Medium networks)
   - Default Mask: 255.255.0.0
3. Class C: 192 – 223 (Small networks)
   - Default Mask: 255.255.255.0
4. Class D: 224 – 239 (Multicast)
5. Class E: 240 – 255 (Reserved)

Private IP Ranges:
- 10.0.0.0 – 10.255.255.255
- 172.16.0.0 – 172.31.255.255

- 192.168.0.0 – 192.168.255.255

[DIAGRAM: IP CLASSES]

Class A: 0xxxxxxx
Class B: 10xxxxxx
Class C: 110xxxxx
Class D: 1110xxxx
Class E: 1111xxxx

(c) DNS [2.5 Marks]

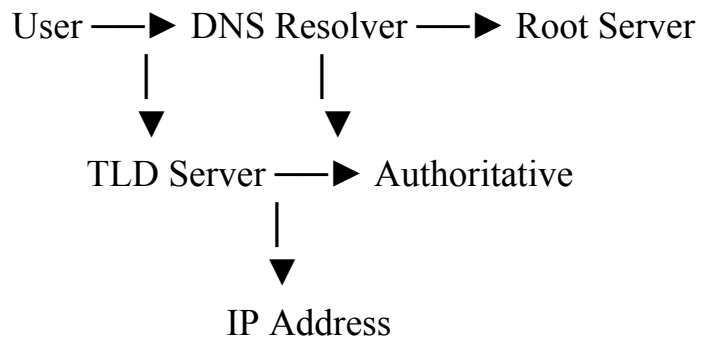
Answer:
DNS (Domain Name System) ek hierarchical naming system hai jo domain names ko IP addresses mein convert karta hai.

Working:
1. User browser mein "google.com" type karta hai.
2. DNS resolver query karta hai.
3. Root Server → TLD Server → Authoritative Server.
4. IP address return hota hai.
5. Browser IP par connect karta hai.

Types of DNS Servers:
1. Recursive Resolver
2. Root Server
3. TLD Server
4. Authoritative Server

Port: 53 (UDP/TCP)

[DIAGRAM: DNS RESOLUTION]



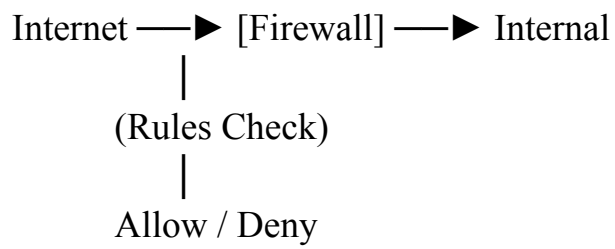
(d) Firewall [2.5 Marks]

Answer:
Firewall ek network security device hai jo incoming aur outgoing traffic ko monitor aur control karta hai.

Types:
1. Packet Filtering: Header check.
2. Stateful Inspection: Connection state track.
3. Application Layer: Data content check.
4. NGFW: Advanced features.

Functions:
- Traffic filtering
- Access control
- Logging
- VPN support

[DIAGRAM: FIREWALL]



(e) Subnetting [2.5 Marks]

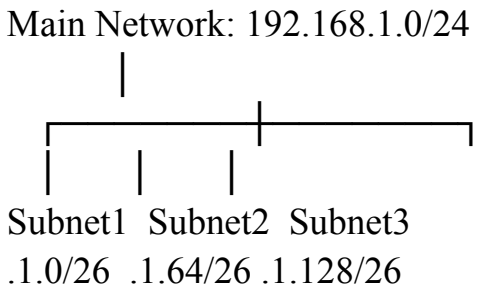
Answer:
Subnetting ek process hai jismein ek bade network ko
chhote sub-networks mein divide kiya jata hai.

Benefits:
1. Network performance improve hoti hai.
2. Security badhti hai.
3. IP addresses efficiently use hote hain.
4. Broadcast traffic kam hota hai.

Subnet Mask:
- 255.255.255.0 (/24)
- 255.255.0.0 (/16)

Example:
Network: 192.168.1.0/24
Subnets: 192.168.1.0, 192.168.2.0, ...

[DIAGRAM: SUBNETTING]



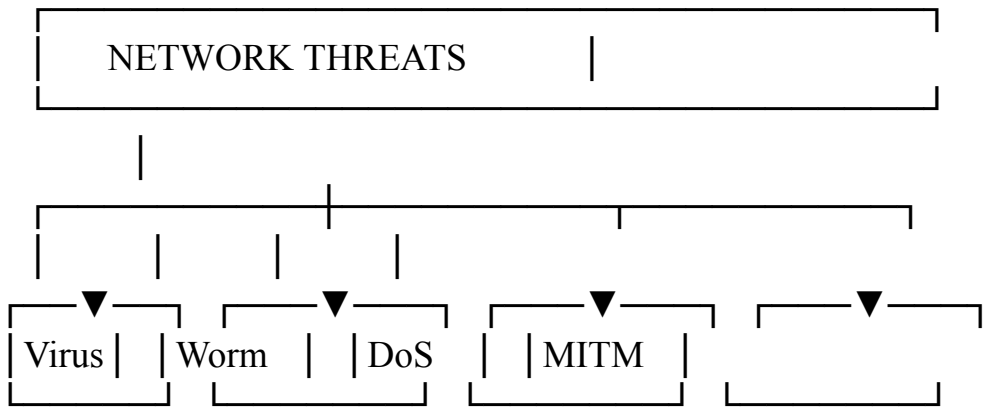
(f) Network Security Threats [2.5 Marks]

Answer:
Network Security Threats wo dangers hain jo network ki security ko compromise karte hain.

Common Threats:
1. Virus: Self-replicating malware.
2. Worm: Network ke through spread.
3. Trojan: Disguised malware.
4. DoS/DDoS: Service disrupt karna.
5. Phishing: Fake emails se info steal.
6. MITM: Data intercept karna.
7. Spoofing: Identity fake karna.
8. SQL Injection: Database attack.

Protection:
- Firewall
- Antivirus
- Encryption
- Strong Passwords
- Regular Updates

[DIAGRAM: THREATS]



END OF NOTES
ALL THE BEST! 🔥
